Solvency Field Theory
V11 Derivation Companion

Part 3

CMB Transfer, Strong-Field Closure, Predictions, and Theorem Ledger

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Purpose. This companion volume supports V11 Master Sections 12–16. The master paper states the early-universe sector, the strong-field closure, the falsification program, and the theorem ledger in scientific-conversational form. This document supplies the derivation skeleton, status distinctions, and mathematical audit trail underneath those claims.

This is a derivation companion, not the narrative master paper. Equations are intentionally centered and prominent. Long reproduction pipelines and code-level audit trails remain separate reproducibility material.

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1 Source hierarchy and status discipline

The companion follows the V11 master structure, but it does not use version chronology as the authority. When older and newer sources differ, the following hierarchy is used.

1. **V11 Master Final Parts 1–3**: current claim and status source.
2. **Repaired Lagrangian documents**: current action source, including the background-relative quadratic potential and exact static vacuum branch collapse.
3. **V10.10 Mesoscale Bridge**: current bottom-up source for particle-to-continuum logic.
4. **V10.5–V10.9 theorem notes**: theorem-gate history and branch-selection receipts.
5. **Original V10 master and companion**: historical derivation archive, used where not superseded.

Status. This part treats the V11 master as the claim source and the repaired-action package as the current action source. Older strong-field and action notes are used as historical support only where they do not conflict with the repaired action.

1.1 Notation

The notation is inherited from Companion Parts 1 and 2. The most important convention remains the separation between Planck's constant and the spatial projector:

$$\boxed{h_{\mathrm{P}} \equiv \text{Planck's constant}}, \qquad \boxed{h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu}. \tag{1}$$

The clock congruence is defined by the total bookkeeping field in the repaired action,

$$\chi(x) = \bar{\chi}(x) + \psi(x), \qquad u^\mu \equiv u^\mu = \frac{\nabla^\mu(\bar{\chi} + \psi)}{\sqrt{-\nabla_\alpha(\bar{\chi} + \psi)\nabla^\alpha(\bar{\chi} + \psi)}}. \tag{2}$$

The projected operation density is

$$\rho_{\mathrm{ops}} = \frac{T_{\mu\nu}u^\mu u^\nu + h^{\mu\nu}T_{\mu\nu}}{h_{\mathrm{P}}}. \tag{3}$$

For a perfect fluid in the clock frame,

$$\rho_{\mathrm{ops}} = \frac{\rho + 3p}{h_{\mathrm{P}}}. \tag{4}$$

2 Claim ledger for Master Sections 12–16

Master section	Load-bearing claim	Companion support block	Status

12	The shatter is a field-theoretic instability controlled by local solvency ratio $\Lambda(x,t)$.	Sections 3–4: Landau free-energy form and effective mass.	Structural / field-theoretic
12	The CMB source carries the operator $k^2/a^2\,\delta\Phi$.	Section 5: gradient term variation and Fourier-space source.	Operator-level
12	CLASS implementation validates compatibility and does not detect a signal.	Section 6: solver consistency ledger.	Solver-compatible
12	$\epsilon_\Phi = 1/(4\pi^2)$ is Onsager-derived at transport level.	Section 7: interface Jacobians and status.	Transport-level
13	Framework-level strong-field closure recovers Schwarzschild and standard TOV as complete-screening limit.	Sections 8–9: multiplicative composition and GR limit equations.	Framework-level
13	Repaired action collapses exactly to Einstein-Hilbert in static vacuum.	Section 10: branch conditions and action reduction.	Exact branch
13	Interior action branch gives modified TOV skeleton.	Sections 11–13: effective density/pressure, regularity, Newtonian limit.	Branch-level
14	SFT makes falsifiable predictions across regimes.	Section 14: prediction-to-failure ledger.	Programmatic
15	Open theorem statuses are distinct and auditable.	Section 15: theorem ledger and remaining work.	Status ledger

3 Local shatter control parameter

Master claim. The early-universe shatter is a local transition driven by the ledger solvency ratio.

The local solvency ratio compares the local information-production rate to the local boundary-service rate:

$$\boxed{\Lambda(x,t) = \frac{\dot{s}(x,t)}{H(t)\,i(x,t)}} \tag{5}$$

where $\dot{s}(x,t)$ is the local information-production rate density and $i(x,t)$ is the local boundary-capacity density available to that region. The factor Hi is the local capacity servicing rate when the horizon-scale bookkeeping channel expands at fractional rate H.

The transition condition is

$$\boxed{\Lambda < 1 \text{ solvent}, \qquad \Lambda = 1 \text{ critical}, \qquad \Lambda > 1 \text{ insolvent}.} \tag{6}$$

The shatter is the early-universe case in which the shared radiation-dominated bookkeeping channel crosses $\Lambda = 1$ and reorganizes into independent clock maintenance.

Status. The local ratio supplies the field-theoretic control parameter. The first-principles derivation of the complete channel histories remains part of Theorem 1, but the local instability structure is fixed once Λ is used as the control variable.

4 Landau-form instability and order parameter

Master claim. When the local solvency ratio crosses unity, the lag field develops a symmetry-breaking instability.

A minimal field-theoretic representation of the shatter uses a Landau free-energy density for the lag order parameter Φ:

$$\boxed{\mathcal{F}(\Phi;\Lambda) = \frac{1}{2}A(1-\Lambda)\Phi^2 + \frac{1}{4}B\Phi^4 + \cdots, \qquad A>0, \quad B>0.} \tag{7}$$

The effective mass is the quadratic coefficient:

$$\boxed{M_{\text{eff}}^2 = A(1-\Lambda).} \tag{8}$$

Thus,

$$M_{\text{eff}}^2 > 0 \quad (\Lambda<1), \qquad M_{\text{eff}}^2 = 0 \quad (\Lambda=1), \qquad M_{\text{eff}}^2 < 0 \quad (\Lambda>1). \tag{9}$$

The equilibrium condition is

$$\frac{\partial \mathcal{F}}{\partial \Phi} = A(1-\Lambda)\Phi + B\Phi^3 = 0. \tag{10}$$

Therefore the stationary points are

$$\Phi = 0, \qquad \Phi^2 = \frac{A}{B}(\Lambda-1). \tag{11}$$

For the insolvent branch $\Lambda > 1$, the nonzero equilibrium amplitude is

$$\boxed{\Phi_{\text{eq}} = \pm\sqrt{\frac{A}{B}(\Lambda-1)} \;\propto\; \sqrt{\Lambda-1}.} \tag{12}$$

This is the standard second-order instability form: a control parameter crosses a critical value, the quadratic mass term changes sign, and the order parameter grows as the square root of the distance from criticality.

Status. This is the field-theoretic scaffold of the shatter. The constants A and B encode the effective stiffness and nonlinear saturation of the lag sector; the normalization chain to observed CMB temperature anisotropy remains open.

5 Perturbative source operator

Master claim. The relic lag perturbation couples to the photon-baryon fluid through a phase-curvature operator proportional to k^2/a^2.

Add the leading spatial-gradient term to the free-energy density on FLRW:

$$\mathcal{F}_{\text{grad}} = \frac{1}{2} K_\Phi \frac{(\nabla\Phi)^2}{a^2}. \tag{13}$$

The variational derivative of this term gives

$$\frac{\delta \mathcal{F}_{\text{grad}}}{\delta\Phi} = -K_\Phi \frac{\nabla^2\Phi}{a^2}. \tag{14}$$

For a Fourier mode

$$\Phi(x,\eta) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, \Phi_k(\eta) e^{ik\cdot x}, \qquad \nabla^2\Phi_k = -k^2\Phi_k. \tag{15}$$

Therefore the source operator becomes

$$\boxed{-K_\Phi \frac{\nabla^2}{a^2}\delta\Phi_k \;\longrightarrow\; K_\Phi \frac{k^2}{a^2}\delta\Phi_k.} \tag{16}$$

The minimal Boltzmann source used in the master paper is the packaged version of this operator:

$$\boxed{S_\ell(k,\eta) = \beta_Q(\eta)\, \frac{k^2}{a^2}\, \delta\Phi(k,\eta).} \tag{17}$$

In a finite transition, the source is further restricted by an instability window:

$$\boxed{S_\ell(k,\eta) = \beta_Q(\eta)\, \frac{k^2}{a^2}\, \delta\Phi(k,\eta)\, W\!\left(\frac{k}{k_c(\eta)}\right).} \tag{18}$$

The cutoff $k_c(\eta)$ is fixed by the balance between the effective mass and the gradient operator. A schematic activation condition is

$$\frac{k^2}{a^2} \lesssim |M_{\text{eff}}^2(\eta)| \qquad \Rightarrow \qquad k \lesssim k_c(\eta) \sim a(\eta)|M_{\text{eff}}(\eta)|. \tag{19}$$

Status. The k^2/a^2 operator is an operator-level match between the shatter free-energy description, the CLASS implementation, and the perturbative sector of the single-field action. Exact normalization into the observed temperature anisotropy remains Theorem 3's open chain.

6 CLASS validation ledger

Master claim. The CLASS implementation establishes compatibility with the CMB at the calibrated physical baseline, not observational detection.

The solver-level implementation tests whether the source can be inserted without corrupting standard photon-baryon physics. The validation ledger is:

Check	Meaning	Status
Zero-source recovery	Setting the SFT source amplitude to zero reproduces the standard ΛCDM temperature spectrum.	Passed
Scaling validation	Varying amplitude, transfer factor, source width, and band position produces the expected response.	Passed
Instability-window confirmation	Only a finite band of k-modes is affected; modes outside the band remain suppressed.	Passed
Acoustic-peak preservation	High-ℓ acoustic structure remains unchanged at the calibrated baseline.	Passed
Low-ℓ compatibility	Calibrated baseline lies far below current low-multipole sensitivity.	Passed

The reported calibrated baseline is

$$\boxed{\Delta\chi^2(\ell \leq 29) = 2 \times 10^{-6}.} \tag{20}$$

This is a compatibility result. It says that the SFT source is small enough, at the calibrated baseline, to remain deeply within Planck-allowed low-multipole variation while preserving the standard high-ℓ acoustic structure.

Status. Solver-validated phenomenology. The implementation demonstrates compatibility and correct source behavior; it does not constitute observational detection of the shatter source.

7 Onsager transfer coefficient

Master claim. The CMB transfer coefficient $\epsilon_\Phi = 1/(4\pi^2)$ is derived at transport level from a minimal near-equilibrium interface law.

At the relic-phase/photon-fluid interface, take a minimal Onsager relation between a generalized flux J_Q and generalized force X_Φ:

$$J_Q = L_{Q\Phi} X_\Phi. \tag{21}$$

The transport coefficient must be expressed in the same variables on both sides of the interface. Two conversion factors appear.

7.1 Phase-counting Jacobian

SFT counts complete cycles, while the field perturbation is naturally angular. The conversion is

$$\boxed{J_{\text{phase}} = \frac{\hbar}{h_{\text{P}}} = \frac{1}{2\pi}.} \tag{22}$$

7.2 Cycle-to-linear interface Jacobian

The relic phase perturbation is angular, while the photon-fluid response is linearized. The minimal isotropic interface conversion contributes a second Jacobian,

$$\boxed{J_{\text{int}} = \frac{1}{2\pi}.} \tag{23}$$

Therefore the effective transfer coefficient is

$$\boxed{\epsilon_\Phi = J_{\text{phase}} J_{\text{int}} = \frac{1}{2\pi}\frac{1}{2\pi} = \frac{1}{4\pi^2}.} \tag{24}$$

The important distinction is that the first Jacobian is a bookkeeping conversion between cycle count and angular phase, while the second is an interface conversion between angular phase and linear response.

Status. Onsager-derived at transport level. The remaining interpretive question is whether this fifth appearance is the same phase-convention $\times$ thermal pair as the first four appearances, or a consistent angular-to-linear extension of the same $4\pi^2$ spine.

8 Framework-level strong-field closure

Master claim. The composition principle forces an exponential metric form, and the asymptotically flat vacuum solution recovers Schwarzschild exactly.

The framework-level argument is independent of the repaired action. It begins with multiplicative composition: local maintenance response factors multiply under nested shells. If a shell contributes incremental maintenance burden $\mathrm{d}U$, the redshift factor changes multiplicatively,

$$\mathrm{d}\ln\mathcal{R} = -2\,\mathrm{d}U. \tag{25}$$

Integrating gives

$$\boxed{\mathcal{R}(r) = e^{-2U(r)}.} \tag{26}$$

The static spherical metric can therefore be written in exponential form,

$$\boxed{\mathrm{d}s^2 = -e^{-2U(r)}\mathrm{d}t^2 + e^{2U(r)}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2.} \tag{27}$$

In the complete-screening exterior vacuum, the asymptotically flat solution selects the Schwarzschild factor,

$$\boxed{e^{-2U(r)} = 1 - \frac{2GM}{r}.} \tag{28}$$

Substitution gives the standard Schwarzschild exterior metric,

$$\boxed{\mathrm{d}s^2 = -\left(1 - \frac{2GM}{r}\right)\mathrm{d}t^2 + \left(1 - \frac{2GM}{r}\right)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2.} \tag{29}$$

Status. Framework-level exact exterior closure. The exponential response is forced by multiplicative composition; the Schwarzschild exterior is the unique static, spherically symmetric, asymptotically flat vacuum solution of the resulting GR exterior.

9 Standard TOV as complete-screening interior limit

Master claim. The framework-level strong-field result recovers standard TOV as the GR/complete-screening interior limit.

In the complete-screening hydrostatic limit, local maintenance burden is absorbed into the GR stress-energy structure. The standard static spherical metric is written

$$\mathrm{d}s^2 = -e^{2\Phi(r)}\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{1 - 2Gm(r)/r} + r^2\mathrm{d}\Omega^2. \tag{30}$$

The GR interior equations are

$$\boxed{\frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2\rho(r),} \tag{31}$$

$$\boxed{\frac{\mathrm{d}\Phi}{\mathrm{d}r} = \frac{G\left[m(r) + 4\pi r^3 p(r)\right]}{r\left[r - 2Gm(r)\right]},} \tag{32}$$

and stress-energy conservation gives

$$\boxed{\frac{\mathrm{d}p}{\mathrm{d}r} = -(\rho + p)\frac{\mathrm{d}\Phi}{\mathrm{d}r}.} \tag{33}$$

Combining the last two equations yields the standard TOV equation,

$$\boxed{\frac{\mathrm{d}p}{\mathrm{d}r} = -(\rho + p)\frac{G\left[m(r) + 4\pi r^3 p(r)\right]}{r\left[r - 2Gm(r)\right]}.} \tag{34}$$

In SFT language, this is the complete-screening interior: the local composition principle admits the GR hydrostatic branch when SFT-specific count/source corrections are absorbed into the local strong-field balance.

Status. Standard TOV is recovered as a framework-level complete-screening limit. This is distinct from the repaired-action modified TOV branch, where count and source sectors remain active in the interior.

10 Action-level exterior collapse

Master claim. The repaired action becomes pure Einstein-Hilbert on the exact background-matched static vacuum branch.

The repaired effective action is

$$\boxed{S_{\rm SFT} = \int \mathrm{d}^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \eta\nabla_\mu(\bar{\chi}+\psi)J^\mu_{\rm ops} + \Lambda_C C_{\rm min} - V_{\rm quad} + \mathcal{L}_m\right].} \tag{35}$$

The repaired potential is

$$\boxed{V_{\rm quad} = V_0\left(e^{-\psi}-1\right)^2.} \tag{36}$$

It satisfies

$$\boxed{V_{\rm quad}\big|_{\psi=0} = 0, \qquad \left.\frac{\partial V_{\rm quad}}{\partial\psi}\right|_{\psi=0} = 0, \qquad \left.\frac{\partial^2 V_{\rm quad}}{\partial\psi^2}\right|_{\psi=0} = 2V_0 > 0.} \tag{37}$$

On the background-matched static vacuum branch,

$$\boxed{\psi = 0, \qquad T_{\mu\nu} = 0, \qquad \theta \equiv \nabla_\mu u^\mu = 0.} \tag{38}$$

Then each non-Einstein sector vanishes:

$$V_{\rm quad} = 0, \qquad \partial_\psi V_{\rm quad} = 0, \tag{39}$$

$$J^\mu_{\rm ops} = 0, \qquad E_\chi = T_{\mu\nu}u^\mu u^\nu = 0, \tag{40}$$

$$H_\chi = \frac{1}{3}\nabla_\mu u^\mu = 0, \qquad C_{\rm min} = H_\chi^2 - \alpha(E_\chi + V_{\rm quad}) = 0. \tag{41}$$

The action therefore reduces to

$$\boxed{S_{\rm vac} = \int \mathrm{d}^4x\sqrt{-g}\,\frac{R}{16\pi G}.} \tag{42}$$

Varying with respect to $g^{\mu\nu}$ gives

$$\boxed{G_{\mu\nu} = 0.} \tag{43}$$

For static, spherically symmetric, asymptotically flat vacuum, the solution is Schwarzschild.

Status. Exact action-level exterior closure on the background-matched static vacuum branch. The fully background-independent definition of $\bar{\chi}$ remains open.

11 Modified TOV branch from the repaired action

Master claim. Inside matter, the repaired action does not reduce to pure GR; it yields a branch-level modified TOV system.

Use static spherical variables

$$\mathrm{d}s^2 = -e^{2\Phi(r)}\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{A(r)} + r^2\mathrm{d}\Omega^2, \qquad A(r) = 1 - \frac{2m(r)}{r}. \tag{44}$$

On the promoted interior branch,

$$\psi(r) = 0, \qquad T^{\mu}_{\ \nu} = \mathrm{diag}[-\rho(r), p(r), p(r), p(r)]. \tag{45}$$

The surviving branch quantities are

$$H_\chi = 0, \qquad V_{\mathrm{quad}} = 0, \qquad E_\chi = \rho(r), \qquad \rho_{\mathrm{ops}} = \frac{\rho(r) + 3p(r)}{h_{\mathrm{P}}}. \tag{46}$$

The tt-side branch load is

$$\boxed{\rho_{\mathrm{eff}}^{(tt)} = (1 + \Lambda_C\alpha)\rho(r).} \tag{47}$$

Therefore the mass equation is

$$\boxed{m'(r) = 4\pi r^2(1 + \Lambda_C\alpha)\rho(r).} \tag{48}$$

The source-sector radial stress is

$$\boxed{T^r_{\ r}\big|_{\mathrm{src}} = -\frac{\eta\kappa}{h_{\mathrm{P}}}e^{-\Phi(r)}\left[\rho(r) + 3p(r)\right].} \tag{49}$$

Using a standard perfect-fluid closure for the matter/count part gives the effective radial pressure

$$\boxed{p_{\mathrm{eff}}^{(r)} = (1 + \Lambda_C\alpha)p(r) - \frac{\eta\kappa}{h_{\mathrm{P}}}e^{-\Phi(r)}\left[\rho(r) + 3p(r)\right].} \tag{50}$$

The standard static spherical lapse equation with $p \to p_{\mathrm{eff}}^{(r)}$ gives

$$\boxed{\Phi'(r) = \frac{m(r) + 4\pi r^3 \left[(1+\Lambda_C\alpha)p(r) - \frac{\eta\kappa}{h_{\rm P}} e^{-\Phi(r)}(\rho(r)+3p(r))\right]}{r\,[r-2m(r)]}.} \tag{51}$$

Assuming separate matter conservation on the branch,

$$\boxed{p'(r) = -(\rho(r)+p(r))\Phi'(r).} \tag{52}$$

Thus the branch-level modified TOV system is

$$\boxed{\begin{aligned} m'(r) &= 4\pi r^2(1+\Lambda_C\alpha)\rho(r), \\ \Phi'(r) &= \frac{m(r) + 4\pi r^3 \left[(1+\Lambda_C\alpha)p(r) - \frac{\eta\kappa}{h_{\rm P}} e^{-\Phi(r)}(\rho(r)+3p(r))\right]}{r\,[r-2m(r)]}, \\ p'(r) &= -(\rho(r)+p(r))\Phi'(r). \end{aligned}} \tag{53}$$

Status. Exact mass equation on the repaired $\psi = 0$ branch; radial pressure and hydrostatic equation are branch-level/effective-fluid closure. Full radial variational closure remains open.

12 Center regularity

Master claim. The modified TOV branch is regular at the stellar center.

Expand the fields near $r = 0$:

$$\rho(r) = \rho_c + \rho_2 r^2 + O(r^4), \tag{54}$$
$$p(r) = p_c + p_2 r^2 + O(r^4), \tag{55}$$
$$\Phi(r) = \Phi_c + \Phi_2 r^2 + O(r^4), \tag{56}$$
$$m(r) = m_3 r^3 + m_5 r^5 + O(r^7). \tag{57}$$

Substitute into the mass equation:

$$3m_3 r^2 + 5m_5 r^4 + \cdots = 4\pi r^2(1+\Lambda_C\alpha)(\rho_c + \rho_2 r^2 + \cdots). \tag{58}$$

Matching powers gives

$$\boxed{m_3 = \frac{4\pi}{3}(1+\Lambda_C\alpha)\rho_c, \qquad m_5 = \frac{4\pi}{5}(1+\Lambda_C\alpha)\rho_2.} \tag{59}$$

The numerator of Φ' is $O(r^3)$ and the denominator is $r[r-2m(r)] = r^2 + O(r^4)$, so

$$\boxed{\Phi'(r) = O(r), \qquad p'(r) = -(\rho+p)\Phi' = O(r).} \tag{60}$$

There are no $1/r$ or constant singular terms at the origin.

Status. Center regularity is branch-level closed for the repaired modified TOV skeleton.

13 Weak-field / Newtonian limit

Master claim. The modified TOV branch has a clean weak-field limit.

Assume

$$\frac{2m}{r} \ll 1, \qquad |\Phi| \ll 1, \qquad p \ll \rho, \qquad e^{-\Phi} \simeq 1. \tag{61}$$

Then the branch system reduces to

$$\boxed{m'(r) \simeq 4\pi r^2 (1 + \Lambda_C \alpha)\rho(r).} \tag{62}$$

The lapse equation becomes

$$\Phi'(r) \simeq \frac{m(r)}{r^2} - 4\pi\Gamma r\rho(r), \qquad \boxed{\Gamma \equiv \frac{\eta\kappa}{h_{\mathrm{P}}}}. \tag{63}$$

The pressure equation is therefore

$$\boxed{p'(r) \simeq -\rho(r)\left[\frac{m(r)}{r^2} - 4\pi\Gamma r\rho(r)\right].} \tag{64}$$

The count sector rescales the enclosed mass, while the source sector adds a local operation-density correction to the radial acceleration.

Status. The weak-field limit is clean and interpretable, but interior phenomenology requires numerical integration with realistic equations of state and comparison to observations.

14 Predictions and falsification ledger

Master claim. Every major sector of the theory gives a concrete way to challenge or kill the framework.

Regime	Prediction	Potential falsifier	Status
Galactic	λ correlates with structural photometry through the Gamma-family morphology curve.	Continuous photometric morphology map fails systematically.	Empirical target

Galactic	High-resolution 21-cm rotation curves follow SFT interpolating function at predicted morphology-dependent λ.	Statistically significant population inconsistent with predicted λ.	SKA/WALLABY target
Cosmological	Growth-rate split between voids and overdensities at $z \sim 0.5$.	Robust null result at predicted precision.	DESI/Euclid target
Cosmological	Supervoid ISW cross-correlation.	Absence of predicted correlation in CMB-S4-quality data.	Future target
Dark energy	$w(z)$ crosses $w = -1$ near $z \approx 0.3$–0.5.	Future data establish constant w with no crossing.	DESI audit target
Strong field	Modified TOV branch implies neutron-star mass-radius deviations.	NICER and future constraints exclude branch predictions.	Open phenomenology
Action	No freely propagating bookkeeping scalar in promoted branch.	Detection of a propagating light scalar with SFT coupling structure.	Branch test
CMB	Completed normalization chain predicts a definite low-ℓ amplitude.	Predicted amplitude above Planck sensitivity but not observed.	Theorem 3 target

Status. Predictions are programmatic at this stage: some already have preliminary support, some are compatibility checks, and some become sharp only after open derivation chains are completed. The falsification ledger is therefore a research map as well as a risk map.

15 Open theorem ledger

Master claim. The V11 frontier is no longer an undifferentiated open problem; it is a set of named closure targets.

Theorem / sector	Current status	What remains
$4\pi^2$ spine (Theorem 2)	Closed at theorem level	Appearance 5 interpretation: same factor or consistent extension
Projection unification (Theorem 5)	Closed at theorem level	No theorem-level remainder
Morphology-to-λ (Theorem 4)	v1 empirical closure	Continuous structural-photometry map to α
Shatter / engine-crash unification (Theorem 1)	Effective closure	First-principles channel histories
CMB transfer (Theorem 3)	Solver-validated phenomenology	Normalization chain from relic phase to observed temperature anisotropy

Beyond FLRW (Theorem 6)	Lead branch selected; exterior gate passed	Interior closure, perturbative normalization, background-independent $\bar{\chi}$

The framework's frontier has shifted from discovery of candidate mechanisms to completion of derivation chains. The surviving structure is now:

$$\boxed{\text{one mechanism} \quad + \quad \text{one source target} \quad + \quad \text{one coefficient spine} \quad + \quad \text{one repaired action program.}} \tag{65}$$

Status. V11 does not claim that every derivation chain has reached the floor. It claims that the remaining floors are named, localized, and testable.

16 Companion Part 3 result ledger

Result	**Equation / object**	**Status**
Local shatter control	$\Lambda(x,t) = \dot{s}/(Hi)$	Control variable
Instability mass	$M_{\text{eff}}^2 = A(1-\Lambda)$	Field-theoretic
Order parameter	$\Phi_{\text{eq}} \propto \sqrt{\Lambda - 1}$	Structural
Boltzmann source operator	$S_\ell = \beta_Q(k^2/a^2)\delta\Phi$	Operator-level
CMB baseline	$\Delta\chi^2(\ell \leq 29) = 2 \times 10^{-6}$	Compatibility
Transfer coefficient	$\epsilon_\Phi = 1/(4\pi^2)$	Onsager-derived
Framework exterior	$e^{-2U} = 1 - 2GM/r$	Exact GR exterior
Action exterior	$S_{\text{vac}} = \int \sqrt{-g}\, R/(16\pi G)$	Exact branch
Modified mass equation	$m' = 4\pi r^2(1 + \Lambda_C \alpha)\rho$	Exact on branch
Modified TOV skeleton	Equations in Section 11	Branch-level
Center regularity	$\Phi' = O(r),\ p' = O(r)$	Branch-level
Weak-field limit	$\Phi' \simeq m/r^2 - 4\pi\Gamma r\rho$	Branch-level

Closing note

This companion volume supports the final third of the V11 master paper. It is intentionally colder than the master. The master says what SFT is trying to say: gravity is what happens when ticks concentrate, and expansion is what happens when they are summed. This companion records the mathematical skeleton behind the early-universe sector, the strong-field claims, the repaired action's exact exterior branch, the modified interior branch, and the theorem ledger.